



Robotics labs

Bachelor's in Data Science and Engineering

2024-2025

Final Assignment

Webots

Professors:

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Introduction

The final assignment of the course can be delivered in groups of two students. The students are expected to achieve a solid expertise using Webots simulator through the following milestones:

- Strong knowledge using robot sensors and actuators.
- Ability to implement basic algorithms related to mobile robots.
- Ability to program a controller to drive a mobile robot through an unexplored map.

Objective

Students will have to design a controller for the Pioneer II robot that allows it to explore the world and find people in distress in an environment that emulates a disaster. The project will have a finite set of test environments similar to the ones used during the labs. The assignment will consist of designing a controller for the Pioneer II to explore the map and find to find two people in danger. The robot should be able to dodge obstacles and walls as it explores the map to rescue the victims. Groups will compete with one another to optimize the time for rescue.

World

- The scenario consists in a rectangular room (10 x 20 m), as shown in Figure 1.
- There are ten versions of the base scenario. They differ in the distribution of the obstacles and the light conditions. Figure 2 shows a scenario with fog and obstacles stacked in three locations of the map.
- Two people will be located at the end of the room waiting to be rescued by Pioneer II. They are represented as green cylinders.
- Students will be provided with some sample worlds to validate their controllers. Professors will choose one of the ten scenarios to test the robot's ability to navigate an unexplored map.

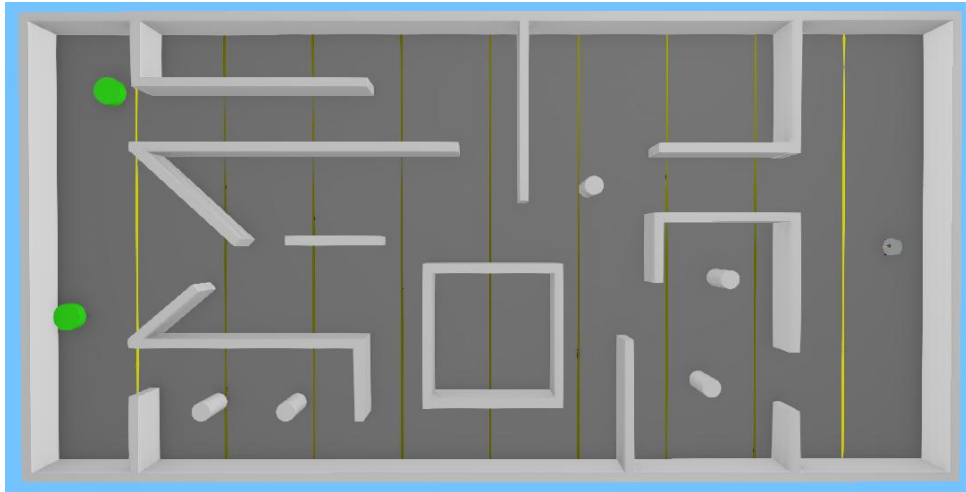


Figure 1. Base scenario



Figure 2. Scenario with fog and structural damage

Robot: Pioneer II

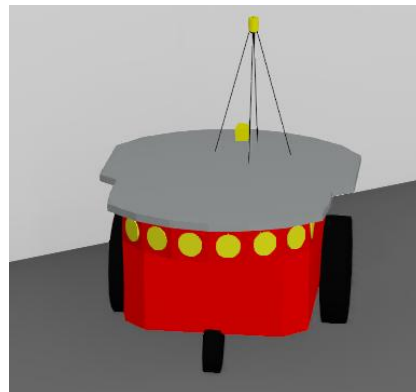
Pioneer II will be used to explore the map and proceed with the rescue. The model of the robot is the same as the one used during the labs and cannot be modified.

Noise has been added to sensor measurements and actuator response to achieve a higher degree of realism in the simulation. Students might use the following sensors: **compass, GPS, distance sensors, encoders and forward and spherical cameras.**

IMPORTANT! The GPS will have a resolution of 3 meters, so it is recommended to use it to correct the accumulated error of other location systems such as odometry, not for real-time robot localization.



Real robot



Simulated robot with two additional cameras

Figure 3. Pioneer II robot

Competition

Objectives:

1. Drive the robot from one end of the scenario to the opposite one, dodging obstacles and walls.
2. Find the two people asking for help (green cylinders)
3. Return to the departure line (yellow line)
4. Optimize the time needed to accomplish the assignments.

Rules:

1. All groups will be provided with the **same scenarios** for validation.
2. Students will **submit their controllers before the deadline**, prior to the competition.
3. Groups will give a **short presentation (5 min)** to show their work and explain their approach. Then, they will test their controller using the **selected scenario**.
4. Robots will have a maximum of 4 minutes to complete the task.
5. Robots will be always located within the departure line at the start. Nevertheless, the **initial position** and **orientation** of the robot, as well as the **selected scenario** for the competition will be chosen randomly before the competition starts.
6. Robots will proceed with the rescue by **stopping within 1 meter to the person and spinning once** to indicate they found the victim. **The terminal will also output a message** when a person is found.
7. Collisions should be minimized. Multiple or big crashes might be penalized.

Grading

Score will be based on:

- Total time to accomplish the complete task
- If task is not completed within the 4 minutes, students will be grade according to:
 - o Total distance traveled in the direction of interest (number of traversed black lines)
 - o Amount of people the robot was able to find.
 - o Total distance traveled towards the departure line (number of traversed black lines)
- Collisions will result in penalties

Evaluation

Students will have to submit the following documents:

- **Report:** You are required to clearly describe the working scheme of the controller implemented in the Pioneer II. List the sensors used and justify the design of the controller using UML diagrams (Information Flow Diagrams, Class Diagrams). Include empirical results like “distance traveled”, “time for rescue” in the diagram (**50 % of the grade**).
- **Presentation:** Students will show their work during class by giving a short presentation (5 min). This will account for **10% of the grade**.
- **Project.** Each group will deliver the source files of the controller and the executable. The organization of the behaviors exhibited by the robot, the performance of the controller and the score obtained in the competition will be evaluated. The quality, comments and structure of the code will also be considered as part of the grade (**40% of the grade**).

Delivery

Each group will submit a ZIP containing the following files:

- Controller source files (.h, .cpp, Makefile), and executable file (.exe)
- Report (.pdf)
- Presentation (.ppt).